Illumination-Prior-Guided Monochrome–Infrared Fusion for Low-Light Aerial Imaging

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***Abstract*—The superior performance of visible–infrared image fusion in ground-level low-light imaging has driven significant advances in this field. However, aerial low-light scenes are characterized by wide illumination variations, complex textures, and motion blur, making it difficult for existing methods to simultaneously restore details, correct illumination, and integrate complementary information. To overcome these challenges, we propose an aerial low-light monochrome–infrared fusion network (ALMIN), which pioneers the use of monochrome–infrared imaging for aerial low-light enhancement. First, we propose an illumination prior extractor that leverages local illumination priors to enhance the texture details of aerial monochrome images, effectively preventing detail loss during subsequent fusion and generating dynamic weights that capture regional illumination variations in aerial scenes. Second, we introduce prior illumination dynamic weights to guide a dual-modality fusion restorer, effectively integrating monochrome and infrared information for the reconstruction of illumination and reflectance components. Within this framework, we design a frequency-refined denoising block (FRDB), a global illumination perception block (GIPB), a fine-grained reflectance perception block (FRPB), and a light-guided fusion block (LGFB) to achieve the effective brightness correction and detail restoration. Third, we construct aerial low-light fusion set (ALFSet), the first multimodal low-light aerial image dataset captured with a monocular and an infrared stereo camera, to evaluate the proposed ALMIN. Extensive quantitative and qualitative experiments demonstrate the superiority and effectiveness of our ALMIN. The source code and dataset will be released on: https://github.com/PeiyangLin1/ALMIN**

***Index Terms*—Low-light aerial imaging, monochrome–infrared image fusion, Retinex theory.**

I. Introduction

AERIAL photography aims to capture ground scenes and moving targets and has been widely applied in fields such as unmanned aerial vehicle (UAV) tracking [1], urban planning [2], and small object detection [3]. However, under low-light conditions, aerial images often suffer from uneven exposure and detail loss, which can hinder subsequent downstream tasks.

For low-light aerial imaging, many existing methods primarily rely on the Retinex theory to restore single-modality visible-light images (see [7], [8], [9], [10]). However, due to the limitations of visible light in low-light conditions, these methods often face inherent constraints. In recent years, due to the illumination-invariant nature of infrared imaging, many studies have leveraged the complementary characteristics of infrared and red, green, and blue (RGB) images to generate well-exposed results (see [4], [5], [6]). Although these methods have achieved remarkable progress on ground-level images with regular structures and uniform exposure, they struggle with the unique challenges of aerial scenarios, as illustrated in Fig. 1(c)–(f). First, low-light aerial images are captured at long distances. This often results in significant noise and numerous small targets whose features are not prominent. Previous methods mainly focus on fusing complementary RGB and infrared features while neglecting the limitations of unimodal feature extraction. This deficiency often leads to direct loss of targets in low-light aerial scenarios. Second, the wide imaging range of aerial images leads to substantial regional illumination variations. Existing methods typically ignore this property and perform feature extraction directly, which can result in the loss of fine-grained detail due to varying illumination. Third, existing RGB–infrared fusion methods are constrained by the inherent limitations of RGB imaging. The presence of the Bayer color filter array reduces the photoelectric efficiency, requiring longer exposure times in low-light conditions. This in turn leads to motion blur and introduces accumulated noise during demosaicing in aerial imaging. These challenges hinder the applicability of existing methods in real-world scenarios, highlighting the necessity of developing networks specifically tailored for low-light aerial imaging.

To address the aforementioned challenges, we construct aerial low-light fusion set (ALFSet), the first multimodal low-light aerial image dataset captured with a monocular and an infrared stereo camera and propose an aerial low-light monochrome–infrared fusion network (ALMIN). Compared with conventional low-light image enhancement using color cameras, the monochrome sensor removes the Bayer filter and eliminates the demosaicing process. Consequently, it achieves higher photoelectric efficiency, captures finer details, and allows shorter exposure times to suppress the motion blur in aerial imagery. As shown in the Fig. 1(d) and (e),

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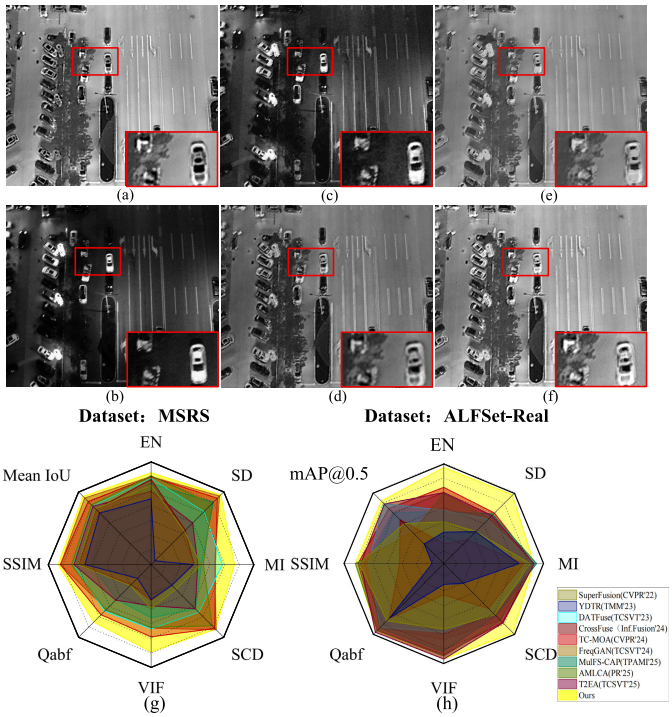


Fig. 1. Our ALMIN achieves state-of-the-art performance in low-light aerial imaging tasks. (a) Low-light aerial infrared image. (b) Low-light aerial monochrome image. (c)–(f) Outputs of our method and comparable methods [4], [5], [6]. (g)–(h) Performance comparison on the ALFSet real-world dataset and the MSRS dataset.

ALMIN achieves state-of-the-art performance on ALFSet, MSRS, and LLVIP datasets. The contributions of this work are summarized as follows.

1) We construct the ALFSet, the first multimodal low-light aerial image dataset captured with a monocular and an infrared stereo camera, comprising both synthetic and real data. This dataset contributes to advancing research and development in the monochrome image domain.
2) We propose an illumination prior extractor that leverages local illumination priors to enhance texture details in aerial monochrome images. Meanwhile, dynamic weights are generated to capture regional illumination variations in aerial scenes, which are further utilized to guide the fusion of monochrome and infrared images.
3) We propose a dual-modality fusion restorer that decomposes monochrome images into illumination and reflection components. The global and local details extracted from infrared images are utilized to restore the illumination and reflection components of monochrome images, respectively. Experimental results demonstrate that ALMIN achieves state-of-the-art performance.

The remainder of this article is organized as follows. Section I reviews related works on low-light aerial image enhancement. Section III presents the motivation of our network and provides detailed information on ALMIN. Section IV validates the effectiveness of ALMIN. Finally, Section V concludes this article.

II. Related Works

A. Low-Light Aerial Image Enhancement

Low-light aerial image enhancement aims to capture the rich target information for downstream tasks. Xi et al. [8] proposed a progressive filtering block that enhanced low-light images by performing adaptive pixel value adjustment via nonuniform illumination estimation. Yao et al. [11] adopted a dual-domain feature fusion network to stagewise restore the brightness and detail information in remote sensing images. Xing et al. [12] leveraged the internal cyclic relationships within remote sensing images to accurately estimate illumination. Lu et al. [9] used an adaptive pre-enhancement block and luminance interval loss to improve the low-light aerial image quality. Lin et al. [13] first introduced the use of a handheld monochrome camera to restore imaging blur in color cameras under low-light conditions. Yuan et al. [7] further leveraged monochrome images to guide the restoration of illumination in RGB images by providing the luminance information in low-light aerial scenarios.

In summary, existing methods rely on visible-light images to estimate illumination and restore details, but they fail to fully recover all information, especially in regions with extreme lighting. Although leveraging monochrome images to guide RGB image restoration achieves promising results, these approaches are still limited to the visible-light spectrum. To address these challenges, we introduce thermal infrared images. These images are illumination-invariant and provide complementary structural and detail information in low-light regions. This method enables more effective low-light image enhancement.

B. Infrared and RGB Image Fusion

In recent years, infrared and RGB image fusion methods include: GAN-based methods, Transformer-based methods, and state-space network (SSM)-based methods. Ma et al. [14] proposed FusionGAN to maintain RGB image details. Fu et al. [15] used densely connected blocks in the generator to better retain texture in fused images. Vs et al. [16] applied Transformers to image fusion, using a multiscale autoencoder and a spatial Transformer strategy to aggregate local and global features. Ma et al. [17] proposed SwinFusion, which leverages Swin Transformers with self- and cross-attention blocks to capture long-range dependencies within and across modalities. Wang et al. [18] leveraged Mamba's long-sequence capability to design an efficient aggregation mechanism for cross-modal image fusion. Lv et al. [19] introduced a structure-aware fusion strategy for low-light infrared and visible images by combining illumination enhancement with edge-based structural modeling. Dong et al. [20] introduced a co-enhancement framework for multimodality image fusion and object detection via feature adaptation.

Although existing RGB–infrared fusion methods achieve promising results in ground-level low-light enhancement by exploiting modality complementarity, their performance is limited in aerial scenarios. RGB images suffer from false color distortion and motion blur caused by low-photoelectric response in low-light environments. These issues restrict the

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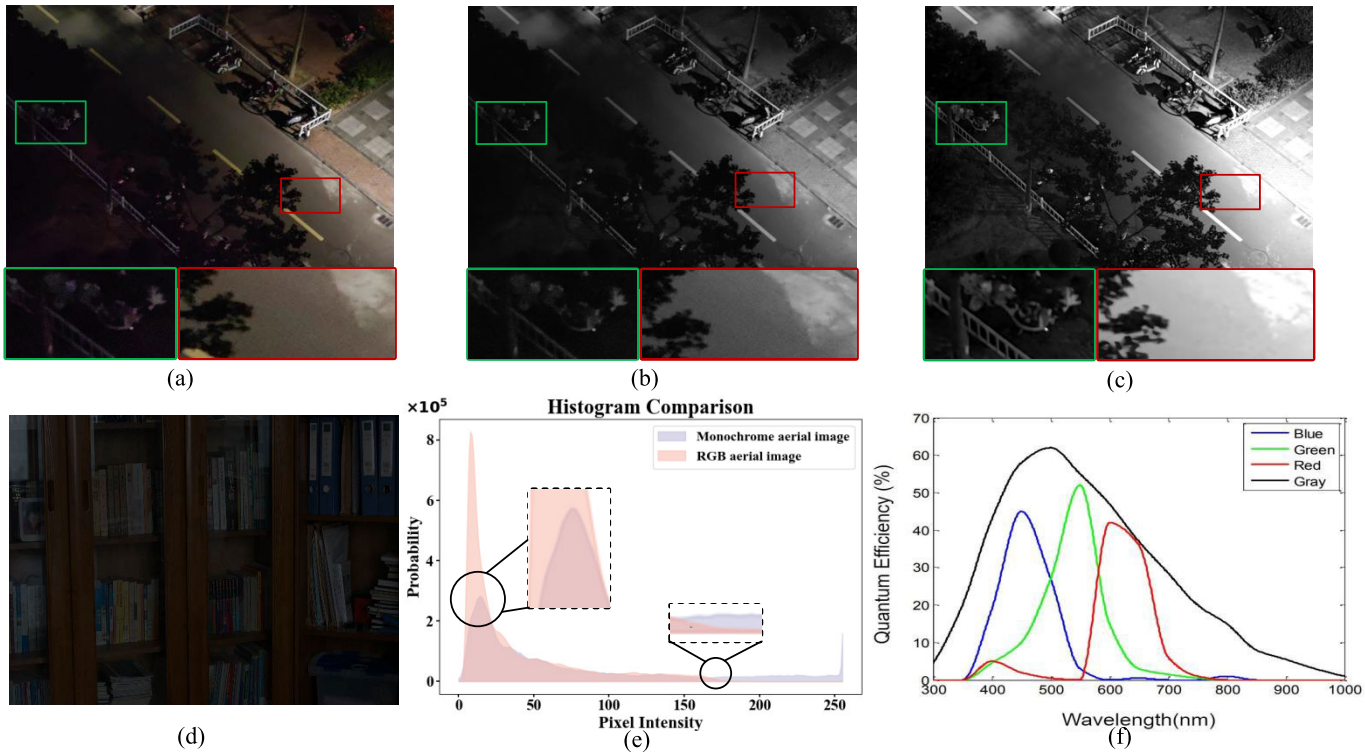


Fig. 2. Comparison of monochrome aerial images, RGB aerial images, and RGB ground-level images. (a) RGB aerial image. (b) Grayscale version of RGB image. (c) Monochrome aerial image. (d) RGB ground-level image. (e) Histogram comparison of RGB and monochrome images. (f) RGB-monochrome camera sensitivity [21].

development of RGB-based fusion methods. Furthermore, aerial images exhibit rich texture details and large dynamic variation ranges. This requires models specifically tailored for aerial scenes. To overcome these limitations, we propose the first monochrome–infrared fusion framework for aerial low-light enhancement. Our approach alleviates illumination interference and enhances fine-grained details before feature fusion.

III. Methodology

A. Problem Formulation

Visible–infrared image fusion aims to generate high-quality, well-exposed results by integrating the complementary information present in each modality. To this end, existing methods typically employ a feature extraction network $\mathcal{E}(\cdot)$ followed by a fusion network $\mathcal{F}(\cdot)$ to extract and combine features from an input visible image I_{low} and infrared image I_T, producing an enhanced counterpart I_{en}

$$I_{\text{en}} = \mathcal{F}\left(\mathcal{E}\left(I_{\text{low}}\right), \mathcal{E}\left(I_T\right); \theta\right) \tag{1}$$

where θ represents a parameter of $\mathcal{F}(\cdot)$.

However, existing visible–infrared fusion networks are primarily developed for ground-level RGB and infrared images. As shown in Fig. 2(a) and (d), aerial images exhibit dense and complex textures as well as significant regional variations in exposure. Existing methods often overlook these unique characteristics and directly perform multimodal feature extraction and fusion, making it difficult to simultaneously achieve detail restoration, illumination correction, and complementary feature integration.

To address the aforementioned challenges, we explicitly network the imaging process. According to the Retinex theory, an image can be decomposed into a detailed reflectance component R and a uniformly exposed illumination component L, as: $I = R \odot L$, where $\odot$ denotes the elementwise multiplication. However, the Retinex network assumes that the observed image I is lossless [22], which is inconsistent with real underexposed aerial scenarios. First, due to the reduced aperture and insufficient light intensity in low-light aerial imaging, camera noise and quantization errors are significantly amplified, resulting in noticeable noise in the reflectance component [7], [22], [23], [24]. Second, uneven illumination distribution may cause local overexposure and underexposure [7], [22], [23]. Third, long-distance imaging inevitably introduces noise as light propagates through the atmosphere, where suspended particles and water droplets cause scattering and attenuation [25], [26]. To more accurately characterize this low-light aerial imaging property, we introduce an illumination perturbation factor $\hat{L}$, a reflectance perturbation factor $\hat{R}$ and a propagation noise factor $\hat{N}$, both correlated with the illumination information of the monochrome image [7], [22], [23], [24], [25], [26]. We reformulate the Retinex theory as follows:

$$I_{\text{low}} = \hat{N} \odot \left(\hat{R} + R\right) \odot \left(\hat{L} + L\right). \tag{2}$$

Inspired by prior works [23], [27], our goal is to obtain a well-exposed image $I_{\text{en}} = R \odot L$. To this end, we construct $\mathcal{F}_R(\cdot)$ and $\mathcal{F}_L(\cdot)$ to extract R and L from the infrared and visible

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Fig. 3. Visualization of aerial image components and modalities. (a) Monochrome aerial image. (b) Infrared aerial image. (c) Initially enhanced aerial image. (d) Illumination aerial image. (e) Histogram distributions before and after enhancement.

images, respectively. These features are then fused to produce the enhanced image I_{en} with the uniform illumination and rich details

$$\begin{aligned} R &= \mathcal{F}_R\left(I_{\text{low}}, I_T; \theta_R\right) \\ L &= \mathcal{F}_L\left(I_{\text{low}}, I_T; \theta_L\right) \\ I_{\text{en}} &= R \odot L. \end{aligned} \tag{3}$$

where θ_R and θ_L are the parameters of $\mathcal{F}_R$ and $\mathcal{F}_L$, respectively.

B. Monochrome and Infrared Image Fusion

First, compared with RGB images, monochrome images I_{mono} dispense with the Bayer color filter array and the demosaicing process. They directly capture incident light, thereby preserving richer detail and texture information. This is illustrated in the green boxes in Fig. 2(a)–(c). In addition, as shown in the red box in Fig. 2(a), the demosaicing process introduces color distortion under low-signal-to-noise ratio conditions. Second, as illustrated in Fig. 2(e), monochrome images exhibit superior contrast compared with RGB images. Third, as shown in Fig. 2(f), monochrome sensors possess a wider spectral response, thus capturing the information across a broader range of wavelengths. Finally, due to the absence of the Bayer filter array, monochrome cameras exhibit higher quantum efficiency, enabling better handling of the tradeoff between motion blur and exposure time in low-light aerial imaging scenarios [13].

Therefore, we propose to use I_{mono} instead of I_{low} in (3) to obtain I_{en} in low-light aerial imaging scenarios. Following (2), we progressively process the monochrome and infrared images to generate I_{en}.

First, aerial images suffer from severe loss of texture details in multimodal fusion tasks. This is caused by insufficient unimodal feature extraction. To mitigate this problem, we leverage a pretrained Retinex decomposition network. This network estimates the illumination component of the input image ($\hat{L} + L$), as illustrated in Fig. 3(d). We then use the estimated illumination $\tilde{L}$ to preliminarily enhance the texture details of low-light aerial monochrome images. As illustrated in the green boxes in Fig. 3(a) and (c), the features in the monochrome images become more prominent following this step. As shown in Fig. 3(e), the enhanced images exhibit superior contrast. This can be mathematically represented as follows:

$$I_P = I_{\text{mono}} \odot \tilde{L}^{-1} = \hat{N} \odot \left(\hat{R} + R\right) \odot \left(L \odot \tilde{L}^{-1} + \hat{L} \odot \tilde{L}^{-1}\right) \tag{4}$$

where I_P denotes a preliminarily brightened image.

Second, to remove the noise introduced during the atmospheric propagation that may affect subsequent processing, we design $\mathcal{D}(\cdot)$ to separate denoising operations for the infrared and monochrome images. This process can be formally expressed as follows:

$$\begin{aligned} \tilde{I}_P &= \mathcal{D}\left(I_P\right) = \left(\hat{R} + R\right) \odot \left(L \odot \tilde{L}^{-1} + \hat{L} \odot \tilde{L}^{-1}\right) \\ \tilde{I}_T &= \mathcal{D}\left(I_T\right). \end{aligned} \tag{5}$$

Third, the direct image restoration often suffers from illumination inconsistencies, leading to the loss of aerial details. To address this, we construct $\mathcal{I}(\cdot)$ and $\mathcal{R}(\cdot)$ to separately extract the illumination component B_P and the reflectance component D_P from the monochrome image. On the other hand, as shown in Fig. 3(a) and (b), monochrome images suffer from severe information loss under extreme illumination, whereas infrared images retain rich texture details due to their illumination-invariant imaging mechanism. Therefore, it is necessary to extract both global B_T and local detail features D_T from the infrared modality. Subsequently, these extracted features are used to restore the reflection and illumination components of the monochrome images. Since B_T exhibit similar characteristics to B_P, while D_T is more consistent with D_P, we also employ $\mathcal{I}(\cdot)$ and $\mathcal{R}(\cdot)$ to extract the global and local infrared features, enabling the efficient feature transfer and reducing model complexity

$$\begin{aligned} B_P &= \mathcal{I}\left(\tilde{I}_P\right) = L \odot \tilde{L}^{-1} + \hat{L} \odot \tilde{L}^{-1} \\ D_P &= \mathcal{R}\left(\tilde{I}_P\right) = \hat{R} + R \\ B_T &= \mathcal{I}\left(\tilde{I}_T\right), \quad D_T = \mathcal{R}\left(\tilde{I}_T\right). \end{aligned} \tag{6}$$

Finally, as shown in Fig. 3(d), $\tilde{L}$ obtained from the pretrained Retinex decomposition network effectively reveals the complementary relationship between the monochrome and infrared modalities. Therefore, we design $\mathcal{K}(\cdot)$, which leverages this component to guide the fusion of monochrome and infrared information. This process can be formally expressed as follows:

$$\begin{aligned} L &= \mathcal{K}\left(B_P, B_T, \tilde{L}\right) \\ R &= \mathcal{K}\left(D_P, D_T, \tilde{L}\right) \\ I_{\text{en}} &= R \odot L. \end{aligned} \tag{7}$$

In summary, our goal is to design $\mathcal{D}(\cdot)$, $\mathcal{R}(\cdot)$, $\mathcal{I}(\cdot)$, and $\mathcal{K}(\cdot)$.

C. Proposed ALMIN

Based on the above analysis, we propoes the ALMIN to obtain I_{en}, as illustrated in Fig. 4. The ALMIN framework

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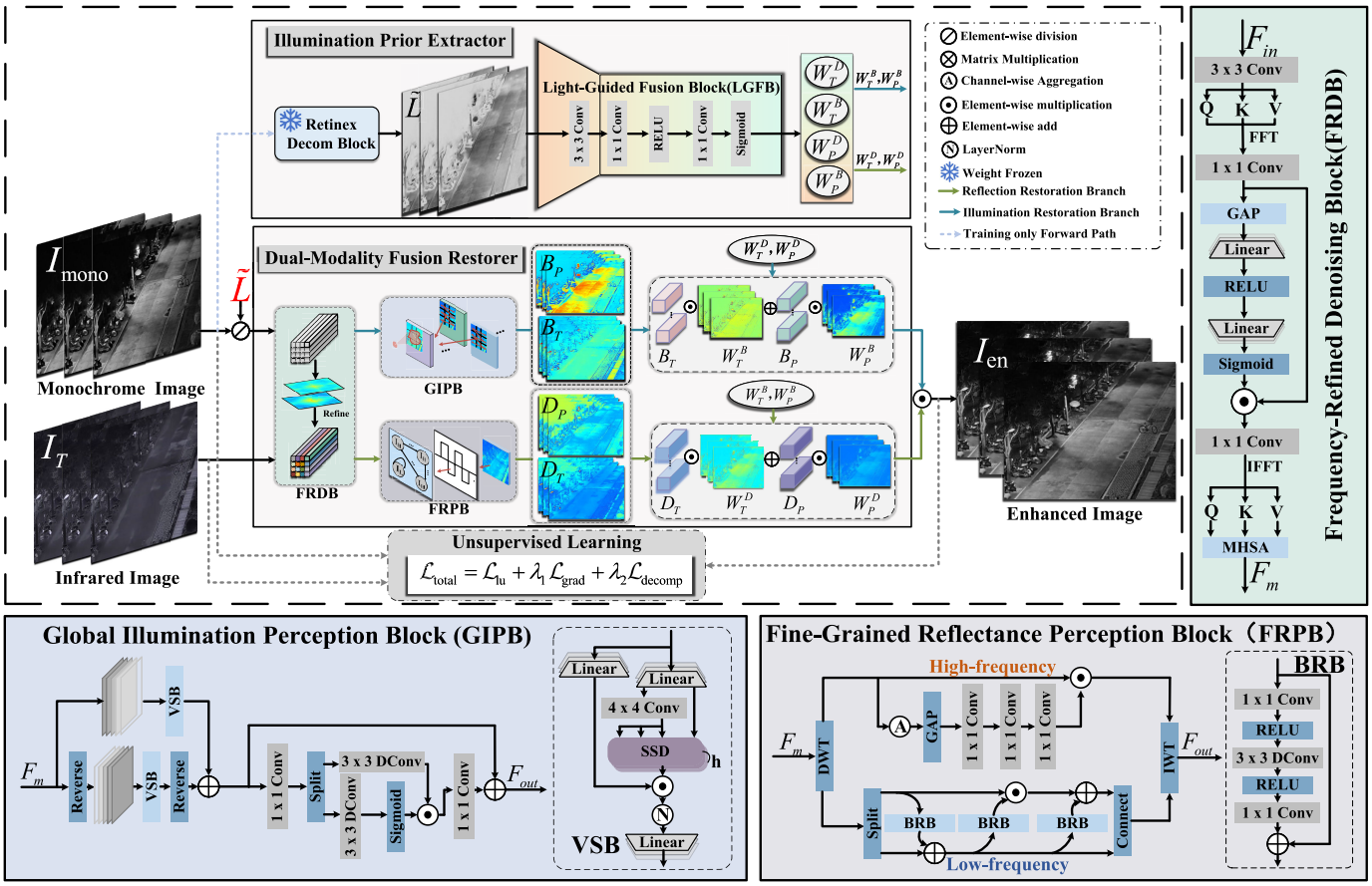


Fig. 4. Architecture of our proposed ALMIN: 1) dual-modality fusion restorer includes our proposed FRDB, GIPB, and FRPB and 2) illumination prior extractor is composed of a Retinex decomposition network and our proposed LGFB.

consists of two main components: the illumination prior extractor and the dual-modality fusion restorer. First, we propose to employ a Retinex decomposition network pretrained on the LOLv2 dataset [28] to extract $\tilde{L}$, and leverage it to preliminarily enhance the texture details of monochrome images. Second, we design a frequency-refined denoising block (FRDB) to perform frequency-domain denoising on both the monochrome and infrared images. Thus, $\mathcal{D}(\cdot)$ is replaced by FRDB. Third, we design a global illumination perception block (GIPB) and a fine-grained reflectance perception block (FRPB) to perform the operations $\mathcal{I}(\cdot)$ and $\mathcal{R}(\cdot)$, respectively. Finally, we propose a light-guided fusion block (LGFB) that leverages $\tilde{L}$ to generate dynamic weights capturing regional illumination variations in aerial scenes. These weights are then used to guide the multimodal feature fusion. Thus, $\mathcal{K}(\cdot)$ is replaced by LGFB.

Dual-Modality Fusion Restorer: To obtain I_{en}, we design the FRDB, GIPB, and FRPB based on (5) and (6). These blocks perform the noise suppression. They also extract the illumination and reflectance components of the monochrome image and the global and local detail features of the infrared image.

1) Frequency-Refined Denoising Block: Low-light aerial images are often affected by noise introduced during atmospheric propagation. However, conventional fusion methods do not account for this characteristic and directly perform feature extraction, which may lead to the information loss. To address this issue, we propose performing denoising on the acquired aerial images using the FRDB. Considering that noise in low-light aerial images tends to concentrate within specific frequency bands, while in the spatial domain it is strongly coupled with image structures, making direct denoising less effective [29]. Therefore, we introduce a frequency-domain dynamic aggregation mechanism to achieve more effective noise suppression. First, we project an input image into Q, K, and V using standard convolutional layers. Second, the Fourier transform is performed on Q, K, and V, mapping them from the spatial domain to the frequency domain. Third, in the frequency domain, we design dynamic filters that perform pointwise convolution and channel aggregation [30] to effectively suppress noise in both the magnitude and phase components. Finally, the inverse Fourier transform is employed to map the frequency-domain representations back into $\tilde{Q}$, $\tilde{K}$, and $\tilde{V}$ for attention score computation. This process is formally defined as follows:

$$\begin{aligned}\left[Q_z, K_z, V_z\right] &= \mathcal{F}_z\left(\left[\text{Conv}_Q, \text{Conv}_K, \text{Conv}_V\right](F_{\text{in}})\right)\\ \left[\tilde{Q}, \tilde{K}, \tilde{V}\right] &= \mathcal{F}_z^{-1}\left(\left[M_f(Q_z), M_f(K_z), M_f(V_z)\right]\right)\\ F_m &= \text{MHSA}\left(\tilde{Q}, \tilde{K}, \tilde{V}\right)\end{aligned} \tag{8}$$

where $M_f(\cdot)$ denotes the dynamic filters, MHSA$(\cdot)$ indicates a multihead self-attention. $F_z(\cdot)$ represents the frequency-domain

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transform, and $F_z^{-1}(\cdot)$ is its inverse, mapping features back to the spatial domain.

2) Global Illumination Perception Block: To effectively extract B_P and B_T from complex aerial images, we propose the GIPB. This block is built upon SSD block [31]. In this block, multihead SSDs perform shared recursive control through the parallel computation to extract B_P and B_T and suppress noise. Meanwhile, a relative position mask L effectively exploits the spatial locality of feature information to enhance feature extraction. However, the causal nature of SSD limits bidirectional information flow. To address this, we perform both forward and backward scans of the token sequence. The scan results are seamlessly fused together. This fusion constructs a complete global context that accommodates the noncausal characteristics of image data. Furthermore, flattening a 2-D feature map into a 1-D sequence disrupts the inherent spatial relationships between image patches. To address this issue, we apply depthwise separable convolutions to the processed 2-D feature maps to explicitly model local neighborhood relationships across feature channels. In addition, we adaptively modulate these local features through a gating mechanism. This design complements global modeling with additional spatial context, thereby ensuring the spatial consistency of B_P and B_T.The process is defined as follows:

$$\begin{aligned} F_G &= L\left(C^T B\right) F_m + \mathrm{Rev}\left(L\left(C^T B\right) \mathrm{Rev}\left(F_m\right)\right) \\ F_{\text{out}} &= \mathrm{GATE}\left(F_G\right) \end{aligned} \tag{9}$$

where

L is defined as follows:

$$L = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ a_1 & 1 & \cdots & 0 \\ a_2 a_1 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{i-1} a_{i-2} \cdots a_1 & a_{i-1} a_{i-2} \cdots a_2 & \cdots & 1 \end{bmatrix} \tag{10}$$

where Rev$(\cdot)$ denotes the reversal operation, GATE$(\cdot)$ indicates the gating mechanism described above, and $C, B \in \mathbb{R}^{P \times N}$ are the learnable parameters. For example, $L_{ij} = a_i a_{i+1} \cdots a_j$ represents the positional decay from a_i to a_j.

3) Fine-Grained Reflectance Perception Block: In low-light images, the rich details in D_P and D_T are often severely corrupted by the interference of ambient illumination. To alleviate this problem, we propose the FRPB. The FRPB employs wavelet decomposition to separate images into illumination-sensitive low-frequency components and high-frequency components that are relatively robust to brightness variations. As illustrated in Fig. 5, by exchanging the low- and high-frequency subbands between real low-light and normal-light images from the DuLAI dataset [7], it can be clearly observed that the low-frequency components contain most of the illumination information, while the high-frequency components mainly preserve edge and structural details that are largely independent of lighting conditions. Motivated by this observation, FRPB adopts distinct feature extraction strategies for the low- and high-frequency components. Considering the instability of single-direction high-frequency information, we first stacks the three high-frequency components along the channel dimension and sums them to obtain a unified feature representation. Next, the global context information is extracted by applying global average pooling. This is followed by channelwise dimensionality reduction through convolution and nonlinear activation. Standard convolution layers are then used to generate attention vectors for each high-frequency component. These attention vectors are multiplied with the original high-frequency components. This process enables both noise suppression and detail enhancement in the high-frequency information. Meanwhile, we leverage an invertible neural network (INN) block [32] for low-frequency components. Its affine coupling layers establish a bidirectional mapping that separates ambient illumination interference and preserves finer details

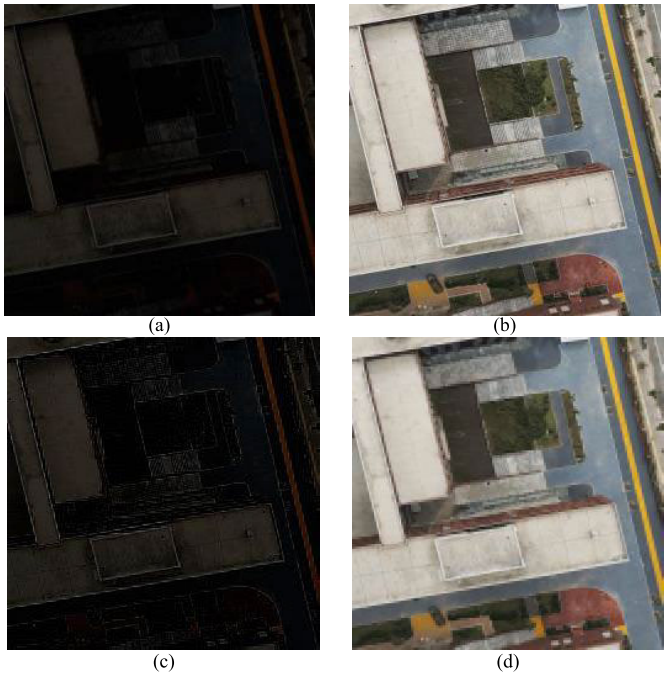


Fig. 5. Motivation for our proposed FRPB. (a) Low-light image. (b) Normal-light image. (c) Image reconstruction using the fusion of low-light low-frequency and normal-light high-frequency components. (d) Image reconstruction using the fusion of low-light high-frequency and normal-light low-frequency components.

$$\begin{aligned} \left[F_{\text{HH}}, F_{\text{HL}}, F_{\text{LH}}, F_{\text{LL}}\right] &= \mathrm{DWT}\left(F_m\right) \\ \left[\tilde{F}_{\text{HH}}, \tilde{F}_{\text{HL}}, \tilde{F}_{\text{LH}}\right] &= \mathrm{Atten}\left(F_{\text{HH}}, F_{\text{HL}}, F_{\text{LH}}\right) \\ \tilde{F}_{\text{LL}} &= \mathrm{INN}\left(F_{\text{LL}}\right) \\ F_{\text{out}} &= \mathrm{IWT}\left(\tilde{F}_{\text{HH}}, \tilde{F}_{\text{HL}}, \tilde{F}_{\text{LH}}, \tilde{F}_{\text{LL}}\right) \end{aligned} \tag{11}$$

where Atten$(\cdot)$ denotes the attention mechanism described above, INN$(\cdot)$ refers to the INN, DWT$(\cdot)$ stands for the discrete wavelet transform, and IWT$(\cdot)$ indicates the inverse wavelet transform.

a) Llumination prior extractor: To achieve the effective fusion of features extracted from infrared and monochrome images, we propose the LGFB. In low-light aerial imagery, the extensive imaging range often results in pronounced spatial variations in illumination, leading to distinctive and complementary characteristics across infrared and monochrome modalities. However, conventional fusion strategies typically

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neglect such spatial heterogeneity of illumination, thus limiting their effectiveness. To address this, we first extract $\tilde{L}$ from a pretrained Retinex decomposition network [22]. Next, we leverage $\tilde{L}$ to generate dynamic weights that capture regional illumination variations in aerial scenes. Specifically, convolutional layers are first applied to align $\tilde{L}$ with the feature channels of both modalities. A lightweight multilevel perception block is then introduced to capture multiscale contextual information, adaptively modeling spatial distributions of brightness and contrast variations. Through illumination-guided design, this block generates fusion weights, which are subsequently normalized to the range $[0, 1]$ via a sigmoid activation. Finally, these weights are employed to perform weighted fusion of the monochrome and infrared information to recover R and L. Based on this, (7) can be expressed as follows:

$$\begin{aligned} W &= \mathcal{K}(I_L) \\ L &= W_T^B \odot B_T + W_P^B \odot B_P \\ R &= W_T^D \odot D_T + W_P^D \odot D_P \\ I_{\text{en}} &= R \odot L \end{aligned} \tag{12}$$

where $W = [W_T^B, W_T^D, W_P^B, W_P^D]$ denotes the set of fusion weights.

b) Loss functions: In unsupervised learning, the loss function plays a crucial role in constraining the network to achieve desirable outcomes.

First, to obtain an enhanced image with uniform exposure, we design an illumination loss function. This loss function is composed of a pixelwise brightness distribution loss $\mathcal{L}_{\text{pixel}}$ and a regional brightness loss $\mathcal{L}_{\text{region}}$. This loss effectively guides the network to suppress overexposed and underexposed areas in low-light aerial images. Specifically, due to the high light sensitivity of monochromatic images, extreme pixel regions often exhibit large uncertainty. To address this, we apply Gaussian weighting W_N to the pixel values of the monochrome image. The weighted monochromatic image is then used together with the infrared image to compute the intensity loss for the fused image. On the other hand, overexposure and underexposure often occur in spatially adjacent regions. To address this issue, we compute the difference between the minimum pixel value of each region in the fused image and its corresponding ideal minimum value. This approach effectively suppresses local dark and bright spots. The overall process is formally defined as follows:

$$\begin{aligned} W_N &= \frac{1}{\sqrt{2\pi}\sigma} \odot \exp\left(-\frac{(I_{\text{mono}} - 0.5)^2}{2\,\sigma^2}\right) \\ \mathcal{L}_{\text{pixel}} &= \|\max(I_T,\ W_N \odot I_{\text{mono}}) - I_{\text{en}}\| \\ \mathcal{L}_{\text{region}} &= \frac{1}{R}\sum_{r=1}^{R}\left|\min_{p\in I_r}(p) - t\right| \\ \mathcal{L}_{\text{lu}} &= \mathcal{L}_{\text{pixel}} + \mathcal{L}_{\text{region}} \end{aligned} \tag{13}$$

where $\|\cdot\|$ represents the euclidean norm, I_r denotes a local patch of I_{en} containing 16 pixels, and t is set to 0.7. It should be noted that the input image is normalized to the range $[0, 1]$.

Second, to preserve more detailed texture information in the images, we introduce a loss $\mathcal{L}_{\text{grad}}$

$$\mathcal{L}_{\text{grad}} = \|\max(\nabla I_{\text{mono}},\ \nabla I_T) - \nabla I_{\text{en}}\| \tag{14}$$

where ∇ denotes the operation of obtaining the image gradient, which is implemented using the Sobel operator.

Third, to guide the network in fully exploiting the complementary information across modalities, we introduce $\mathcal{L}_{\text{decomp}}$ [33]

$$\mathcal{L}_{\text{decomp}} = \frac{(\text{CC}(D_P, D_T))^2}{\text{CC}(B_P, B_T) + \varepsilon} \tag{15}$$

where $\text{CC}(\cdot,\cdot)$ computes the correlation coefficient between two inputs, and ε takes the value of 1.01 to ensure this term remains strictly positive.

Finally, the overall loss $\mathcal{L}_{\text{total}}$ for ALMIN is defined as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{lu}} + \lambda_1 \mathcal{L}_{\text{grad}} + \lambda_2 \mathcal{L}_{\text{decomp}} \tag{16}$$

where λ_1 is set to 10 and λ_2 is set to 2.

IV. Experiments

A. Aerial Low-Light Fusion Set

Due to the lack of publicly available paired monochrome and infrared datasets for low-light aerial imagery, we develop the ALFSet consisting of both synthetic and real-world samples for validation purposes.

1) Real-World Dataset: We build a stereo imaging system comprising a Hikvision microthermal camera and a Sony monochrome camera for monochrome–infrared capture. However, because of the resolution gap between the two cameras and the intrinsic disparity of the stereo configuration, it is necessary to perform the image registration on the captured images. Considering that feature points in nighttime aerial images are generally inconspicuous and that significant differences exist between infrared and monochrome modalities. Meanwhile, conventional manual registration methods [34] often lead to low accuracy and high labor costs. To address the above issues, we develop a novel dataset registration method. First, the XoFTR network [35] is pretrained for image feature-point matching on multiview RGB images. Second, fine-tuning is performed on randomly selected pairs of RGB and infrared images. Third, this pretrained network is used to detect matching feature points in unregistered pairs of monochrome and infrared images. Finally, the obtained matching points are used to estimate the homography matrix. The estimation is performed using the RANSAC algorithm. The resulting homography matrix is then applied to perform a perspective transformation on the monochrome images.

2) Synthetic Dataset: We utilize the well-exposed images from the DroneVehicle dataset [36] to generate low-light monochrome and low-light RGB images. The generation is performed using our simulation degradation algorithm. By pairing these images with the original infrared images in the dataset, we construct both low-light monochrome–infrared and low-light RGB–infrared datasets. First, to generate low-light RGB images, we employ a pretrained TPSeNCE network [37] to simulate illumination variations under low-light conditions.

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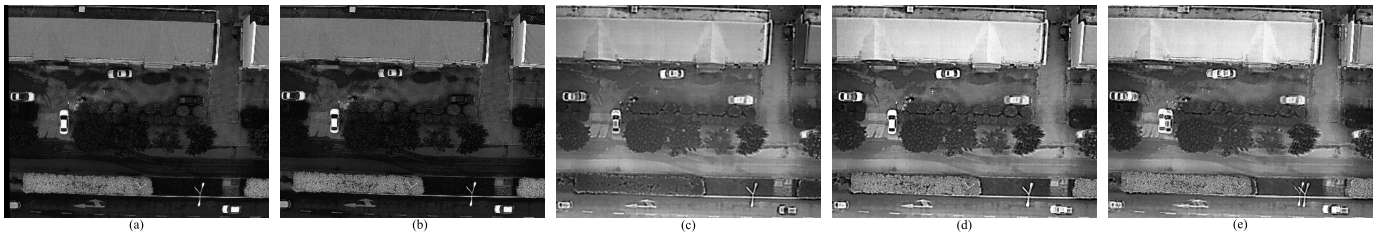


Fig. 6. Comparison of our registration method with DroneVehicle. (a) Our registered monochrome aerial image. (b) Registered monochrome aerial image by DroneVehicle. (c) Infrared aerial image. (d) Image fused after registration using our method. (e) Image fused after registration using DroneVehicle.



Fig. 7. Synthetic dataset validation analysis. (a) Real low-light RGB image. (b) Real normal-light RGB image. (c) Synthetic low-light RGB image. (d) Synthetic low-light monochrome image.

Second, to mimic the noise characteristics of low-light aerial images, we add a Gaussian–Poisson mixed noise. The Poisson component models photon shot noise, while the Gaussian component accounts for residual static noise. On the other hand, to generate low-light monochrome images, we extract the Y channel from the normal-light images and apply a combination of linear and gamma adjustments. Then, we add Gaussian–Poisson mixed noise with the same noise coefficients as used for the low-light RGB images.

3) Data Verification: To validate the effectiveness of our proposed registration method for low-light aerial monochrome infrared datasets, we apply it to the ALFSet-Syn dataset. As shown in Fig. 6, compared with the original DroneVehicle dataset registration, our method achieves precise spatial alignment of the images. This alignment is more conducive to subsequent image fusion. Furthermore, to validate the effectiveness of our proposed low-light simulation algorithm, we perform a visualization analysis. Specifically, we adjust the camera's ISO settings to capture real low-light and normal-light images. These normal-light images are then used to simulate monochrome low-light degradation algorithms and color low-light degradation algorithms. As shown in Fig. 7, the synthetic low-light monochrome and RGB images are visually similar to the real low-light images, while exhibiting clear visual differences from the normal-light images. These results demonstrate the effectiveness of our low-light simulation algorithm.

TABLE I
INFORMATION ABOUT THE ALFSET

ALFSet	Dataset	Number of image pairs	Resolution of image pairs
Real-world	Monochrome Training Set	1200	640×512
	Infrared Training Set	1200	640×512
	Monochrome Test Set	400	640×512
	Infrared Test Set	400	640×512
Synthetic	Monochrome Training Set	1800	640×512
	RGB Training Set	1800	640×512
	Infrared Training Set	1800	640×512
	Monochrome Test Set	900	640×512
	RGB Test Set	900	640×512
	Infrared Test Set	900	640×512

4) Information About ALFSet: The ALFSet consists of aerial images captured under low-light conditions such as dusk and nighttime, from various altitudes and viewing angles. The dataset contains complex scenes featuring elements such as buildings, vehicles, and roads. The ALFSet-Syn Dataset contains object detection labels sourced from the DroneVehicle dataset. The ALFSet-Syn Dataset contains object detection labels sourced from the DroneVehicle dataset [36]. The information about the ALFSet is shown in Table I.

The real-world subset comprises 1600 paired infrared and monochrome images, while the synthetic subset contains 2700 paired infrared, monochrome, and RGB images. The RGB images are further employed in subsequent experiments. The purpose of these experiments is to verify the superiority of fusing monochrome and infrared images. This superiority is demonstrated over traditional RGB–infrared image fusion under low-light conditions. All data are divided into training and testing sets with a ratio of 3:1.

B. Experimental Settings

1) Datasets: We conduct both qualitative and quantitative experiments on the ALFSet dataset. Furthermore, to verify the superiority of our network in nighttime RGB–infrared image fusion tasks, we conduct experiments on nighttime images from the LLVIP [34] and MSRS [38] datasets.

2) Evaluation Metrics: Following [39], we employ seven quantitative metrics to evaluate the fusion performance: entropy (EN), standard deviation (SD), mutual information (MI), sum of correlation of differences (SCD), visual information fidelity (VIF), quality aware based fusion (Qabf), and

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TABLE II

PERFORMANCE COMPARISON ON BENCHMARK DATASETS. RED AND BLUE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Methods	ALFSet-Real Dataset						
	EN (↑)	SD (↑)	MI (↑)	SCD (↑)	VIF (↑)	Qabf (↑)	SSIM (↑)
SuperFusion (JAS'22)	7.00	41.09	1.99	1.26	0.64	0.35	0.95
YDTR (TMM'23)	6.90	40.99	1.41	0.24	0.34	0.27	0.39
DATFuse (TCSVT'23)	7.11	47.48	**2.39**	1.22	0.66	0.52	0.88
CrossFuse (Inf. Fusion'24)	6.79	36.89	1.65	1.06	0.41	0.29	0.64
TC-MoA (CVPR'24)	**7.33**	**50.38**	2.17	1.51	0.81	0.61	0.96
FreqGAN (TCSVT'24)	7.28	48.98	2.19	1.44	0.71	0.52	0.91
MulFS-CAP (TPAMI'25)	7.10	49.96	**2.41**	1.32	0.79	0.54	0.88
AMLCA (PR'25)	7.27	47.67	2.21	1.45	0.82	0.61	**0.98**
T2EA (TCSVT'25)	7.28	49.14	2.20	**1.53**	**0.84**	**0.66**	0.94
Ours	**7.52**	**59.45**	2.18	**1.86**	**0.84**	**0.63**	**0.98**
Methods	**LLVIP Dataset**						
	EN (↑)	SD (↑)	MI (↑)	SCD (↑)	VIF (↑)	Qabf (↑)	SSIM (↑)
SuperFusion (JAS'22)	7.16	42.35	1.86	1.23	0.68	0.35	0.84
YDTR (TMM'23)	6.24	27.77	1.81	0.89	0.45	0.20	0.53
DATFuse (TCSVT'23)	7.09	**49.97**	2.40	1.30	0.56	0.44	0.59
CrossFuse (Inf. Fusion'24)	5.75	20.70	1.78	0.62	0.56	0.30	0.57
TC-MOA (CVPR'24)	**7.35**	46.37	2.52	**1.59**	**0.97**	**0.70**	**0.92**
FreqGAN (TCSVT'24)	7.23	43.00	**2.57**	1.50	0.76	0.48	0.88
MulFS-CAP (TPAMI'25)	6.89	35.62	1.61	1.17	0.34	0.28	0.52
AMLCA (PR'25)	7.07	39.24	2.40	1.45	0.68	0.56	0.85
T2EA (TCSVT'25)	7.10	39.40	2.36	1.45	0.65	0.57	0.78
Ours	**7.40**	**50.62**	**2.82**	**1.84**	**0.84**	**0.67**	**0.93**
Methods	**MSRS Dataset**						
	EN (↑)	SD (↑)	MI (↑)	SCD (↑)	VIF (↑)	Qabf (↑)	SSIM (↑)
SuperFusion (JAS'22)	5.38	20.37	1.35	0.68	0.55	0.23	0.88
YDTR (TMM'23)	4.46	9.41	1.14	0.57	0.44	0.15	0.73
DATFuse (TCSVT'23)	5.85	27.64	**1.94**	1.31	0.81	0.51	0.87
CrossFuse (Inf. Fusion'24)	5.78	20.67	1.16	0.57	0.45	0.22	0.37
TC-MoA (CVPR'24)	5.95	32.56	1.75	1.75	**0.91**	**0.55**	**0.93**
FreqGAN (TCSVT'24)	5.97	**33.36**	1.72	**1.79**	0.80	0.49	0.87
MulFS-CAP (TPAMI'25)	5.51	31.04	1.37	1.22	0.67	0.38	0.72
AMLCA (PR'25)	**6.02**	30.49	1.42	1.68	0.79	0.50	0.92
T2EA (TCSVT'25)	4.96	23.40	1.27	1.21	0.52	0.39	0.34
Ours	**6.23**	**33.80**	**2.27**	**1.77**	**1.10**	**0.63**	**0.98**

structural similarity index measure (SSIM). Higher values of these metrics indicate better fusion quality.

To further demonstrate the effectiveness of our network on downstream object detection tasks, we evaluate two settings. Specifically, we perform object detection using YOLOv8-OBB on the ALFSet-Syn and LLVIP datasets, with mAP@0.5 and mAP@0.5:0.90 as evaluation metrics.

3) Comparison With Current Methods: To evaluate the performance of the proposed network, we compare ALMIN with state-of-the-art networks, including: SuperFusion [40] (JAS'22), YDTR [41] (TMM'23), DATFuse [42] (TCSVT'23), CrossFuse [43] (Inf.Fusion'24), TC-MOA [44] (CVPR'24), FreGAN [45] (TCSVT'24), MulFS-CAP [4] (TPAMI'25), AMLCA [5] (PR'25), and T2EA [6] (TCSVT'25). All comparison methods are trained under the same datasets and experimental settings for a fair evaluation.

4) Implementation Details: We implement ALMIN in PyTorch and the networks are trained using the Adam optimizer. The number of epochs, batch size, and initial learning rate are set to 20, 8, and 1×10^{-4}, respectively. Patch sizes are set to 128×128 for all datasets. All experiments are conducted on a Linux server equipped with an NVIDIA GeForce GTX 4090 graphics card.

C. Comparison on Visual Quality

1) Quantitative Analysis: Table II reports the quantitative results of eight evaluation metrics for various methods. Our proposed ALMIN achieves optimal performance across multiple metrics on the ALFSet real-world dataset. For instance, EN surpasses the second-best network by 0.19, while SD improves by 9.07 over the runner up. This demonstrates that compared to existing low-light image enhancement networks, our network can better extract and fuse detail features from aerial images. Moreover, on RGB–infrared datasets including MSRS and LLVIP, our network also attains the competitive performance across multiple metrics, further validating its generalization capability. While our model's advantages in SD metric are smaller on the ground-level LLVIP and MSRS datasets, they become significantly greater on the aerial ALFSet-Real dataset. This discrepancy confirms that our model is specifically optimized for texture-rich scenarios: it effectively enhances and preserves detailed information in such contexts without unnecessarily intervening in texture-sparse scenes.

2) Qualitative Analysis: Fig. 8 shows the results on the ALFSet real-world dataset. As mentioned in Section I, aerial images have wide imaging coverage and large dynamic range. As shown in the green boxes in Fig. 8, existing methods ignore this characteristic of aerial images. Consequently, they struggle to balance detail recovery in overexposed and underexposed regions, resulting in various degrees of image blur in overexposed regions. As stated in Section I, objects in aerial images are small in size but rich in detail. Existing methods ignore this characteristic of aerial images and directly perform feature extraction and fusion. As illustrated in the red boxes in Fig. 8, this approach leads to detail loss of individual modalities during the multimodal image fusion. In comparison, our proposed network leverages prior illumination information to guide separate restoration of the illumination and reflectance components. It successfully fuses monochrome and infrared images, producing images with the superior visual quality that achieve both uniform exposure and rich detail preservation.

D. Downstream Object Detection

1) Quantitative Analysis: Table III reports the evaluation metrics of images generated by our model and competing methods on the downstream object detection task. The experimental results show that our model achieves superior detection performance on both the ALSEt-Syn and LLVIP datasets, with improved mAP@0.5 and mAP@0.5:0.95 over multiple object categories. These results further confirm that our generated images effectively enhance the performance of downstream tasks.

2) Qualitative Analysis: Fig. 9 demonstrates that YDTR, CrossFuse, TC-MOA, and AMLCA produce apparent false positives resulting from improper enhancement of fine structures, while DATFuse, MulFS-CAP, and SuperFusion exhibit evident false negatives. Although FreGAN and T2EA show fewer false detections, their inadequate restoration of fine details still causes missed truck detections (light blue boxes). In contrast, our approach precisely reconstructs fine details and achieves the robust target detection.

E. Ablation Study

Ablation studies are conducted to test the effectiveness of our proposed networks on the ALFSet real-world dataset. EN,

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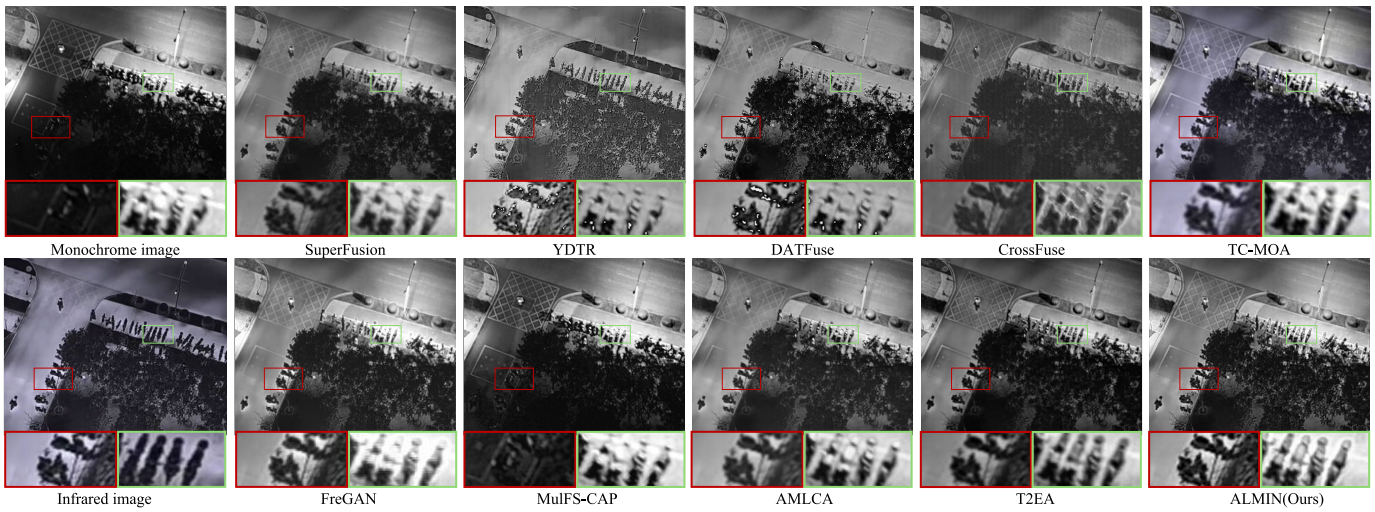


Fig. 8. Qualitative comparison results on the ALFSet real-world dataset.

TABLE III

PERFORMANCE COMPARISON ON ALSET-SYN AND LLVIP DATASETS. RED AND BLUE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Methods	ALSEt-Syn Dataset										LLVIP Dataset			
	mAP@0.5					mAP@0.5:0.95					mAP@0.5		mAP@0.5:0.95	
	Avg	Bus	Car	Freight	Truck	Avg	Bus	Car	Freight	Truck	Avg	Person	Avg	Person
SuperFusion (JAS'22)	0.681	0.682	0.964	0.513	0.564	0.579	0.609	0.813	0.453	0.441	0.974	0.974	**0.789**	**0.789**
YDTR (TMM'23)	0.640	0.644	0.950	0.453	0.512	0.561	0.595	0.795	0.422	0.433	0.951	0.951	0.698	0.698
DATFuse (TCSVT'23)	**0.722**	**0.758**	0.965	0.506	**0.658**	0.616	0.634	0.814	0.474	0.545	0.976	0.976	0.768	0.768
CrossFuse (InfFusion'24)	0.689	0.725	0.964	0.451	0.599	0.602	0.617	0.839	0.457	0.494	0.956	0.956	0.718	0.718
TC-MoA (CVPR'24)	0.712	0.743	0.967	0.491	0.646	0.627	0.660	0.836	0.482	0.533	0.976	0.976	0.783	0.783
FreqGAT (TCSVT'24)	0.704	0.744	0.967	0.482	0.622	0.615	**0.670**	0.834	0.463	0.493	**0.982**	**0.982**	0.764	0.764
MulFS-CAP (TPAMI'25)	0.703	0.723	**0.968**	**0.482**	0.639	0.633	0.658	**0.847**	0.475	**0.552**	0.969	0.969	0.738	0.738
AMLCA (PR'25)	0.664	0.668	0.963	0.446	0.581	0.585	0.615	0.825	0.400	0.498	**0.982**	**0.982**	0.774	0.774
T2EA (TCSVT'25)	0.721	**0.749**	0.967	0.519	0.648	**0.633**	**0.668**	0.826	**0.506**	0.542	0.980	0.980	0.781	0.781
Ours	**0.727**	**0.749**	**0.969**	**0.537**	**0.655**	**0.635**	0.649	**0.853**	**0.483**	**0.545**	**0.983**	**0.983**	**0.782**	**0.782**

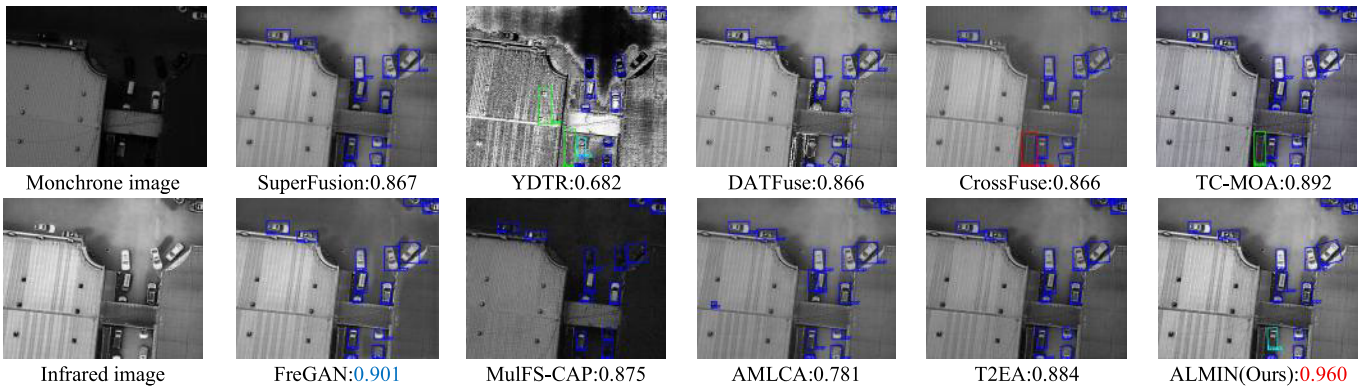


Fig. 9. Qualitative comparison of detection in the ALFSet-Syn dataset. The detection performance is evaluated by mAP@0.5.

SD, SCD, and SSIM are employed to quantitatively evaluate the fusion performance.

1) Ablation Study of Different Blocks: To further validate the superiority of the proposed blocks, we conduct ablation experiments on the GIPB, FRPB, and LGFB, respectively. As shown in Table IV, the proposed GIPB effectively alleviates the limitations of Mamba2 in handling 2-D image data. This demonstrates its superior capability in global modeling compared with the Transformer architecture. In addition, the experiments reveal that the proposed FRPB captures richer image details. This is compared with the configuration employing only the INN block for local feature extraction. Finally,

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Fig. 10. Comparison of object detection results between monochrome–infrared and RGB–infrared image fusion. (a) Infrared image. (b) Monochrome image. (c) RGB image. (d) RGB–infrared image fusion. (e) Monochrome–infrared image fusion.

TABLE IV
ABLATION STUDY OF DIFFERENT BLOCKS

	Configurations	EN (↑)	SD (↑)	SCD (↑)	SSIM (↑)
I	BiMamba2 → Transformer	7.44	53.49	1.65	0.96
II	GIPB → Mamba2	7.22	49.11	1.39	0.78
III	FRPB → INN block	7.40	52.70	1.59	0.95
IV	w/o LGFB	7.38	50.25	1.56	0.82
	Ours	**7.52**	**59.45**	**1.86**	**0.98**

TABLE V
ABLATION STUDY ON THE COMPOSITION OF THE LOSS FUNCTION

	Configurations	EN (↑)	SD (↑)	SCD (↑)	SSIM (↑)
I	w/o $\mathcal{L}_{\text{lu}}$	7.24	51.64	1.52	0.85
II	w/o $\mathcal{L}_{\text{grad}}$	7.06	49.03	0.72	0.44
III	w/o $\mathcal{L}_{\text{decomp}}$	7.24	51.24	1.48	0.86
	Ours	**7.52**	**59.45**	**1.86**	**0.98**

the results confirm the effectiveness of the LGFB. Specifically, it enhances the overall fusion performance.

2) Ablation Study on the Composition of the Loss Function: To verify the effectiveness of the proposed loss function, we conduct ablation experiments by removing $\mathcal{L}_{\text{lu}}$, $\mathcal{L}_{\text{grad}}$, and $\mathcal{L}_{\text{decomp}}$, respectively. As shown in Table V, each loss term contributes to better guiding the fusion of monochrome and infrared images to generate well-exposed and richly detailed low-light aerial images.

3) Ablation Study of Monochrome–Infrared and RGB–Infrared Image Fusion: To evaluate the advantages of monochrome images in low-light aerial and infrared image fusion, we conduct fusion training using paired datasets from the ALFSet-Syn dataset, including monochrome–infrared and RGB–infrared image pairs. The fused images are then used to train a YOLOv8-OBB network for object detection. The experimental results show that images fused with monochrome–infrared achieve a mAP@0.5 of 0.679, whereas RGB–infrared fusion yields a mAP@0.5 of 0.648. As illustrated in Fig. 10, in low-light conditions, RGB images suffer from severe loss of detail. They also exhibit more pronounced local bright and dark regions. These characteristics make fusion with RGB images more prone to false positives and false negatives in object detection.

F. Limitations and Future Work

Our proposed ALMIN framework is the first to perform low-light aerial image fusion using a monochrome–infrared camera pair. However, practical deployment on UAV edge devices requires the model to be more lightweight and real-time. In future work, we will investigate model compression and acceleration techniques. These techniques will enable efficient inference on edge platforms. We will also incorporate graph neural networks to enhance the model's robustness in diverse environments.

V. CONCLUSION

In this article, we propose ALMIN, a novel network for enhancing low-light aerial images by fusing monochrome and infrared inputs. Different from conventional methods, ALMIN first employs an illumination prior extractor. This extractor leverages local illumination priors to enhance the texture details of aerial monochrome images and to generate dynamic weights. Then, guided by these weights, the dual-modality fusion restorer integrates the reliable illumination and reflectance components derived from the monochrome image with the global and detail features from the infrared image. This integration produces uniformly exposed illumination and detailed reflectance components. In addition, we introduce the ALFSet, which contains both synthetic and real-world image pairs. Extensive experiments demonstrate that ALMIN delivers superior performance and visual quality for low-light aerial monochrome–infrared image fusion. In addition, it exhibits strong generalization on ground-level RGB–infrared fusion tasks.

REFERENCES

[1] S. Li, X. Yang, X. Wang, D. Zeng, H. Ye, and Q. Zhao, "Learning target-aware vision transformers for real-time UAV tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4705718.

[2] H. Jung, H.-S. Choi, and M. Kang, "Boundary enhancement semantic segmentation for building extraction from remote sensed image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5215512.

[3] B. Wan, R. Cong, X. Zhou, H. Fang, Y. Sun, and S. Kwong, "RDNet: Region proportion-aware dynamic adaptive salient object detection network in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5647912.

[4] H. Li, Z. Yang, Y. Zhang, W. Jia, Z. Yu, and Y. Liu, "MulFS-CAP: Multimodal fusion-supervised cross-modality alignment perception for unregistered infrared-visible image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 5, pp. 3673–3690, May 2025.

[5] D. Wang et al., "AMLCA: Additive multi-layer convolution-guided cross-attention network for visible and infrared image fusion," *Pattern Recognit.*, vol. 163, pp. 111468–111478, Jul. 2025.

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[6] Z. Huang et al., "T^{2}EA: Target-aware Taylor expansion approximation network for infrared and visible image fusion," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 5, pp. 4831–4845, May 2025.
[7] P. Yuan, L. Lin, J. Lin, Y. Liao, and T. Zhao, "Low-light aerial imaging with color and monochrome cameras," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5619510.
[8] Y. Xi, W. Jia, Q. Miao, J. Feng, J. Ren, and H. Luo, "Detection-driven exposure-correction network for nighttime drone-view object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5605014.
[9] W. Lu, L. Zhu, and S.-B. Chen, "Unsupervised ultra-high-resolution UAV low-light image enhancement: A benchmark, metric and framework," 2025, *arXiv:2509.01373*.
[10] C. Guo et al., "Zero-reference deep curve estimation for low-light image enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 1780–1789.
[11] Z. Yao, G. Fan, J. Fan, M. Gan, and C. L. Philip Chen, "Spatial–frequency dual-domain feature fusion network for low-light remote sensing image enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4706516.
[12] L. Xing, H. Qu, S. Xu, and Y. Tian, "CLEGAN: Toward low-light image enhancement for UAVs via self-similarity exploitation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610714.
[13] J. Lin, S. Pei, B. Chen, N. Jiang, W. Gao, and T. Zhao, "LDRM: Degradation rectify model for low-light imaging via color-monochrome cameras," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 8406–8414.
[14] J. Ma, H. Zhang, Z. Shao, P. Liang, and H. Xu, "GANMcC: A generative adversarial network with multiclassification constraints for infrared and visible image fusion," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–14, 2021.
[15] Y. Fu, X.-J. Wu, and T. Durrani, "Image fusion based on generative adversarial network consistent with perception," *Inf. Fusion*, vol. 72, pp. 110–125, Aug. 2021.
[16] V. Vs, J. M. J. Valanarasu, P. Oza, and V. M. Patel, "Image fusion transformer," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2022, pp. 3566–3570.
[17] J. Ma, L. Tang, F. Fan, J. Huang, X. Mei, and Y. Ma, "SwinFusion: Cross-domain long-range learning for general image fusion via Swin transformer," *IEEE/CAA J. Autom. Sinica*, vol. 9, no. 7, pp. 1200–1217, Jul. 2022.
[18] Y. Wang et al., "MambaPro: Multi-modal object re-identification with mamba aggregation and synergistic prompt," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2025, vol. 39, no. 8, pp. 8150–8158.
[19] G. Lv, X. Gao, A. Dong, Z. Wei, and J. Cheng, "SLFusion: A structure-aware infrared and visible image fusion network for low-light scenes," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 12, pp. 11877–11892, Dec. 2025.
[20] A. Dong et al., "Co-enhancement of multi-modality image fusion and object detection via feature adaptation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 12624–12637, Dec. 2024.
[21] H.-G. Jeon, J.-Y. Lee, S. Im, H. Ha, and I. S. Kweon, "Stereo matching with color and monochrome cameras in low-light conditions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4086–4094.
[22] Y. Cai, H. Bian, J. Lin, H. Wang, R. Timofte, and Y. Zhang, "Retinexformer: One-stage retinex-based transformer for low-light image enhancement," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 12470–12479.
[23] Y. Zhang, J. Zhang, and X. Guo, "Kindling the darkness: A practical low-light image enhancer," in *Proc. 27th ACM Int. Conf. Multimedia (ACM MM)*, Oct. 2019, pp. 1632–1640.
[24] H. Li, X. Hu, and H. Wang, "Interpretable unsupervised joint denoising and enhancement for real-world low-light scenarios unsupervised joint denoising and enhancement for real-world low-light scenarios," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025, pp. 69282–69295.
[25] S. G. Narasimhan and S. K. Nayar, "Chromatic framework for vision in bad weather," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 1, Jun. 2000, pp. 598–605.
[26] S. G. Narasimhan and S. K. Nayar, "Contrast restoration of weather degraded images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 25, no. 6, pp. 713–724, Jun. 2003.
[27] M. Zhou, X. Wu, X. Wei, T. Xiang, B. Fang, and S. Kwong, "Low-light enhancement method based on a retinex model for structure preservation," *IEEE Trans. Multimedia*, vol. 26, pp. 650–662, 2024.
[28] W. Yang, S. Wang, Y. Fang, Y. Wang, and J. Liu, "From fidelity to perceptual quality: A semi-supervised approach for low-light image enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3060–3069.
[29] H. Xia, S. Montresor, R. Guo, J. Li, and P. Picart, "Optimal processing scheme for restoration of phase data corrupted by strong decorrelation noise and dislocations," *Appl. Opt.*, vol. 58, no. 34, pp. G187–G196, 2019.
[30] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[31] T. Dao and A. Gu, "Transformers are SSMS: Generalized models and efficient algorithms through structured state space duality," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 235, 2024, pp. 10041–10071.
[32] M. Zhou, X. Fu, J. Huang, F. Zhao, A. Liu, and R. Wang, "Effective pan-sharpening with transformer and invertible neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5406815.
[33] Z. Zhao et al., "CDDFuse: Correlation-driven dual-branch feature decomposition for multi-modality image fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 5906–5916.
[34] X. Jia, C. Zhu, M. Li, W. Tang, and W. Zhou, "LLVIP: A visible-infrared paired dataset for low-light vision," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 3489–3497.
[35] Ö. Tuzcuoğlu, A. Köksal, B. Sofu, S. Kalkan, and A. A. Alatan, "XoFTR: Cross-modal feature matching transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 4275–4286.
[36] Y. Sun, B. Cao, P. Zhu, and Q. Hu, "Drone-based RGB-infrared cross-modality vehicle detection via uncertainty-aware learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 10, pp. 6700–6713, Oct. 2022.
[37] S. Zheng, C. Lu, and S. G. Narasimhan, "TPSeNCE: Towards artifact-free realistic rain generation for deraining and object detection in rain," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 5382–5391.
[38] L. Tang, Q. Yan, X. Xiang, L. Fang, and J. Ma, "C2RF: Bridging multi-modal image registration and fusion via commonality mining and contrastive learning," *Int. J. Comput. Vis.*, vol. 133, no. 8, pp. 5262–5280, Aug. 2025.
[39] J. Ma, Y. Ma, and C. Li, "Infrared and visible image fusion methods and applications: A survey," *Inf. Fusion*, vol. 45, pp. 153–178, Jan. 2019.
[40] L. Tang, Y. Deng, Y. Ma, J. Huang, and J. Ma, "SuperFusion: A versatile image registration and fusion network with semantic awareness," *IEEE/CAA J. Autom. Sinica*, vol. 9, no. 12, pp. 2121–2137, Dec. 2022.
[41] W. Tang, F. He, and Y. Liu, "YDTR: Infrared and visible image fusion via Y-shape dynamic transformer," *IEEE Trans. Multimedia*, vol. 25, pp. 5413–5428, 2023.
[42] W. Tang, F. He, Y. Liu, Y. Duan, and T. Si, "DATFuse: Infrared and visible image fusion via dual attention transformer," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 7, pp. 3159–3172, Jul. 2023.
[43] H. Li and X.-J. Wu, "CrossFuse: A novel cross attention mechanism based infrared and visible image fusion approach," *Inf. Fusion*, vol. 103, Mar. 2024, Art. no. 102147.
[44] P. Zhu, Y. Sun, B. Cao, and Q. Hu, "Task-customized mixture of adapters for general image fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 7099–7108.
[45] Z. Wang, Z. Zhang, W. Qi, F. Yang, and J. Xu, "FreqGAN: Infrared and visible image fusion via unified frequency adversarial learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 1, pp. 728–740, Jan. 2025.



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